



National University of Computer & Emerging Sciences - NUCES - Karachi

FAST School of Computing

Object-Oriented Programming CL-1004 – LAB 6 TASKS

Task 1: Beam Search with Dynamic Beam Width

PROBLEM:

Modify the Beam Search algorithm so that the **beam width automatically increases if the search is not progressing toward the goal.**

REQUIREMENTS

1. Use the provided graph or create a larger graph with at least 15 nodes.
2. Implement Beam Search with the following behavior:
 - Start with beam width = 2.
 - If the goal is not found after 3 levels, increase the beam width by 1.
 - Maximum beam width allowed = 5.
3. Track and print:
 - Beam nodes at each level.
 - Current beam width.
 - Final path and total cost.

Task 2: Hill Climbing with Random Restart for N-Queens

PROBLEM:

Implement **Random Restart Hill Climbing** to solve the **N-Queens problem for $N = 8$ or $N = 10$.**

REQUIREMENTS

1. Use the standard state representation (list of column positions).
2. Implement:
 - Conflict calculation function
 - Neighbor generation
 - Hill Climbing search
3. Add Random Restart:
 - If Hill Climbing gets stuck in a local optimum, restart with a new random state.
4. Limit the number of restarts to 20 attempts.



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Task 3: Genetic Algorithm for Timetable Optimization

PROBLEM: Design a Genetic Algorithm to generate a class timetable for a small department.

PROBLEM CONSTRAINTS

- 5 teachers
- 5 courses
- 5 time slots per day
- 5 days per week

CONSTRAINTS:

1. A teacher **cannot teach two classes at the same time.**
2. Each course must be scheduled **exactly 3 times per week.**
3. No teacher should teach **more than 3 consecutive classes.**

REQUIREMENTS

1. Represent the timetable as a chromosome.
2. Implement:
 - Population initialization
 - Fitness function (penalty-based)
 - Selection
 - Crossover
 - Mutation
3. Run the algorithm for 200 generations.